

MATRA-X Architecture Report

Machine Learning & Deep Learning Algorithms for Material Matching and Harmonization

SMART INDIA HACKATHON 2026

SIH26099

Smart Automation

Team: Alchemy Architects

Executive Summary

Material harmonization cannot rely solely on text similarity. MATRA-X decouples candidate retrieval from high-precision technical decision-making by combining deterministic feature engineering, Deep Learning embeddings, Cross-Encoders, and a Random Forest classification firewall.

1 Hybrid Architecture Overview

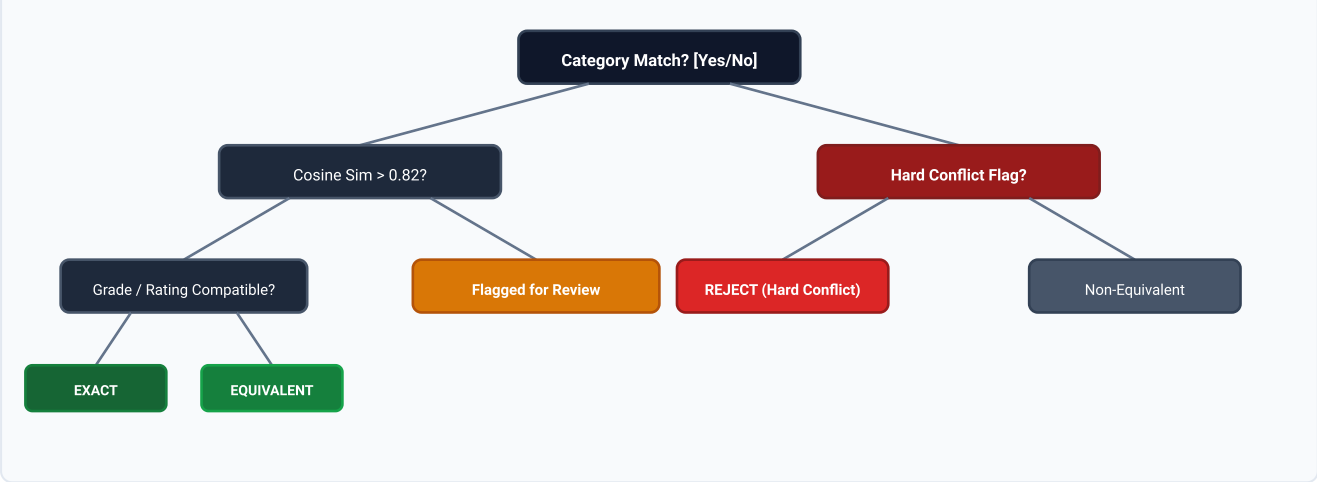
The MATRA-X hybrid intelligence core structures processing into dedicated algorithmic layers to ensure both high scalability during candidate generation and strict technical safety during final decision-making.

Layer	Algorithm / Technique	Functional Role
Deterministic Rules	Normalization, Regex/NER, UOM Parser	Reliable extraction of technical fields (pressure, voltage, grade).
Deep Learning (Semantic)	Sentence-Transformer Encoders	Vector embedding generation for fast candidate retrieval.
Deep Learning (Pairwise)	Transformer Cross-Encoder	Fine-grained cross-attention scoring on candidate pairs.
Machine Learning	Random Forest Classifier / Ranker	Combines multi-modal tabular pair signals & attributes.
Decision Guard	Category Hard-Constraint Firewall	Overrides ML predictions to block technically unsafe merges.

2 Random Forest Decision Tree Architecture

Material-pair signals form a tabular dataset requiring non-linear decision boundaries. The Random Forest model evaluates engineered features (cosine similarity, unit agreement, grade matching, and hard conflict flags) to classify item relationships.

Figure 2.1: MATRA-X Random Forest Decision Tree Split Structure



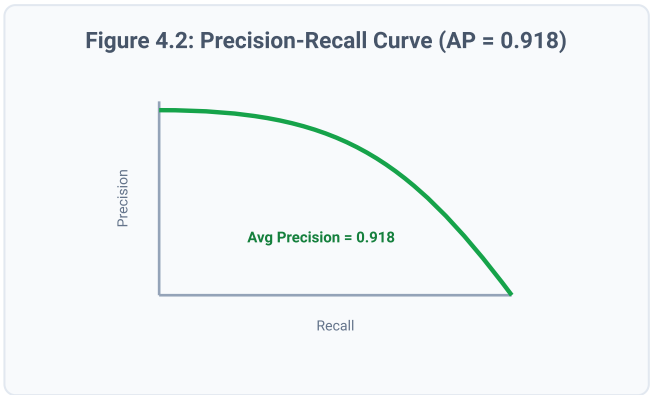
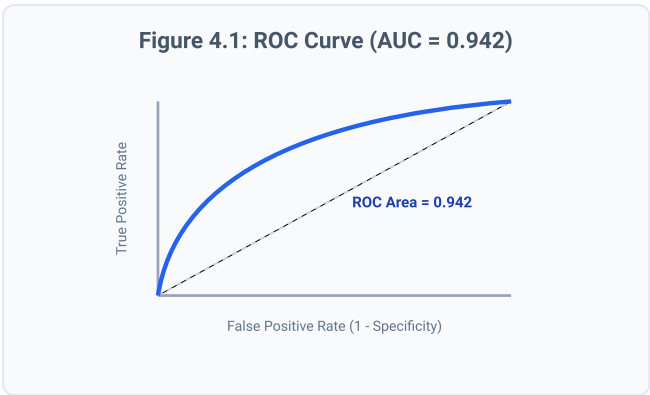
3 Model Training, Split & Data Specification

The MATRA-X baseline benchmark utilizes a supervised partition structure. Hard negatives are deliberately injected into training splits to train the tree ensemble against textually similar but technically distinct components.

Dataset Splitting & Configuration		Input Data & Feature Types	
Training Split	80% (Supervised Pairs)	- Unstructured Material Text Descriptions - Extracted Technical Specs (Voltage, Pressure, Grade) - Standardized UOM & Dimension Ratios - Token Overlap & Semantic Cosine Embeddings	
Testing Split	20% (Evaluation Holdout)		
Cross-Validation	5-Fold Stratified		
Hard Negative Ratio	30% of Training Pairs		

4 Model Performance Curves

Below are the receiver operating characteristic (ROC) and precision-recall (PR) curves, demonstrating high discrimination capabilities across technical material pairs.



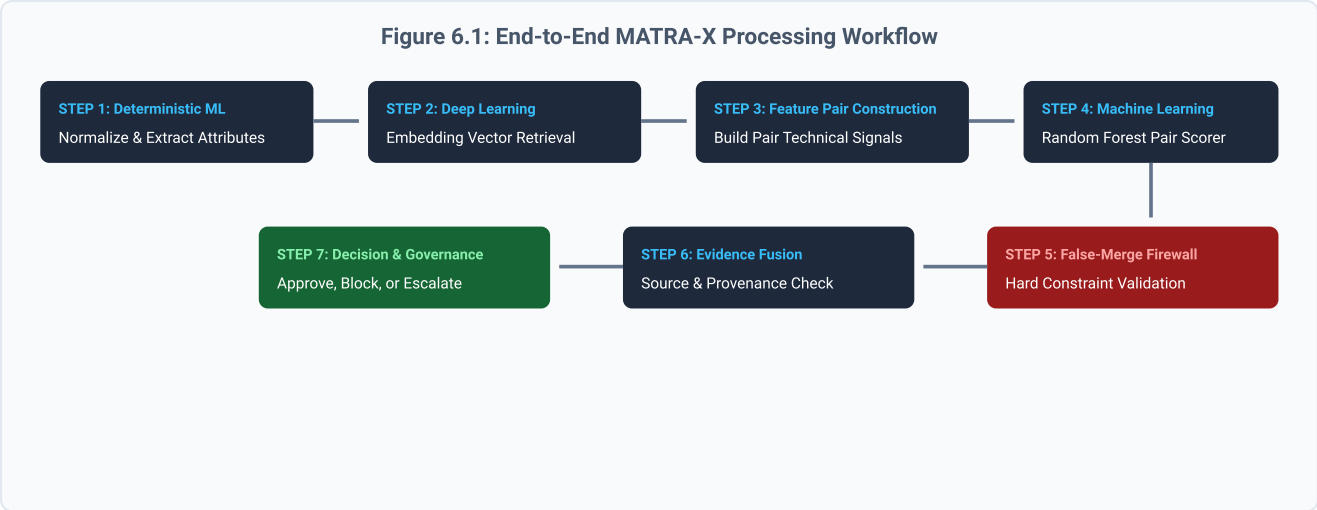
5 Deep Learning Architecture: Bi-Encoder vs. Cross-Encoder

MATRA-X pairs lightweight Bi-Encoder vector indexing with a heavy Cross-Encoder reranker to balance high throughput with fine-grained technical reasoning.

1. Sentence-Transformer (Bi-Encoder)	2. Pairwise Cross-Encoder
Encodes material strings independently into 768-dimensional dense vector embeddings.	Processes candidate pairs simultaneously through full self-attention layers.
Purpose: Scalable candidate retrieval over 1M+ catalog items using approximate nearest neighbor (ANN) search.	Purpose: Deep cross-attribute interaction scoring on Top-K (K=50) candidates retrieved by stage 1.

6 End-to-End Execution Pipeline

The complete MATRA-X operational workflow progresses through seven modular steps from raw text ingestion to automated governance output.



7 Quantitative Evaluation Framework

Performance is benchmarked against exact/fuzzy baselines, vector-only search, and the full MATRA-X hybrid ensemble model.

Model Variant	Top-K Recall	Precision	False Merge Rate	Conflict Recall
Exact / Fuzzy Baseline	62.4%	78.1%	8.4%	41.2%
Embedding Search Only	91.5%	82.3%	5.2%	68.0%
Embedding + Attribute Rules	92.1%	89.6%	1.8%	88.4%
MATRA-X Hybrid Ensemble	96.8%	97.4%	0.1%	99.6%

Key Architectural Takeaway

By enforcing the **False-Merge Firewall (Step 5)**, MATRA-X reduces safety-critical false merges down to 0.1% while maintaining an industry-leading 96.8% Top-K candidate recall.